

CompTIA A+ Core 1 Playbook

IT Labs

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Troubleshoot Power Supply Problems

You've received a service request from an employee experiencing a critical issue where their system shuts down unexpectedly before it can fully boot. You believe the power supply unit (PSU) may be at fault. If it does need to be replaced, select one that includes PCIe connectors for future hardware updates.

A quick reference card is included in the exhibits to explain how to use the Power Supply Tester.

Your task is to test the existing PSU and, if needed, select a compatible replacement that meets both current and future needs. To complete this task, do the following:

- Attempt to boot the computer to verify the issue, and then complete these steps (Instructions 1-2 on quick reference card):
 1. Place the **Power Supply Tester** on the workbench.
 2. Unplug the computer.
 3. Disconnect power to internal hard drives
- Test the power supply (instructions 3-6 on quick reference card):
 1. Transfer the **Connector, Power Supply, 20+4 pin**, and **Connector, Power Supply, 4-pin processor** to the power supply tester.
 2. Attach a **Connector, SATA 15-pin** power connector to the power supply tester.
 3. Plug the computer in and view the Power Supply Tester results.
- Unplug the power cable from the computer and move the bad power supply to the workbench next to the tester. Leave the cables connected to the tester.
- Select and install a new power supply:
 1. Select a power supply from the inventory that has the same 20+4 and 4-pin connectors and the 6-pin PCIe connectors.
 2. Reconnect the **Connector, Power Supply, 20+4 pin** and **Connector, Power**

Supply, 4-pin to the motherboard and **Connector, SATA 15-pin** to each drive.

- Plug the computer in, turn on the power supply, and then start the computer.

Complete this lab as follows:

1. Verify that the computer does not boot and prepare to test the power supply.
 1. Select the **power button** on the computer to verify that it fails to boot.
 2. Select **OK** to close the message.
 3. Under Inventory, expand **Tools**.
 4. Drag the **Power Supply Tester** from the Inventory to the Workspace area.
 5. Above the computer, select the change view icon, then select **Back** to view the back of the computer.
 6. Drag the **power cord** from the computer's power supply to the Workspace.
 7. Above the computer, select the change view icon, then select **Drive Bays**.
 8. Drag the **power connector** (top left) from the optical drive to the Workspace.
 9. Drag the **power connector** (bottom left) from the SATA drive to the Workspace.
2. Test the power supply.
 1. Above the computer, select the change view icon, then select **Motherboard**.
 2. Select the **power supply**. Notice the two cables connected from the power supply to the motherboard are highlighted.
 3. From the motherboard, drag the **Connector, Power Supply, 20+4 pin** power supply connector to the far-right port on the power supply tester.
 4. From the motherboard, drag the **Connector, Power Supply, 4-pin** CPU power connector to the 4/6/8 (upper-left) port on the power supply tester.
 5. From the Selected Item pane, drag an unconnected **Connector, SATA 15-pin** power connector to the bottom-left port on the power supply tester.
 6. Above the computer, select the change view icon, then select **Back**.
 7. Select the *Partial Connections icon* for the wall plate, then select the **AC power cable**.
 8. From the Selected Item pane, drag the **AC Power Connector (Female)** to the power supply port.
 9. Zoom in for better viewing and use the information in the exhibits to determine if the power supply is faulty. In this case, blinking voltage lights, low voltage readings, and low level (LL) indicate it is faulty.
3. Remove the power supply.
 1. If you are not already viewing the back of the computer, select the change view

- icon, then select **Back**.
 2. Drag the **power cord** from the computer's power supply to the Workspace.
 3. Above the computer, select the change view icon, then select **Motherboard**.
 4. Drag the bad **Power Supply** to the Workspace area. Leave the cables connected to the tester.
4. Install a power supply and reconnect power to the motherboard and SATA drives.
 1. Under Inventory, select **Tools** to return to the main Inventory list.
 2. Expand **Power Supplies**.
 3. Drag the **Power Supply, ATX, 20+4 pin, PCIe** power supply to the correct area in the computer.
 4. Verify that the new power supply is selected.
 5. From the Selected Item pane, drag the **Connector, Power Supply, 20+4 pin** power supply connector to the motherboard port.
 6. From the Selected Item pane, drag the **Connector, Power Supply, 4-pin** processor power connector to the motherboard port.
 7. Above the computer, select **Drive Bays** to switch to the drive bays view of the computer.
 8. Select the *Partial Connections icon* above the computer, then select the **Power Supply**.
 9. From the Selected Item pane, drag a **Connector, SATA 15-pin**, to the power port on the optical drive.
 10. From the Selected Item pane, drag a **Connector, SATA 15-pin**, to the power port on the hard drive.
 5. Provide power to the computer and start the computer.
 1. Above the computer, select the change view icon, then select **Back**.
 2. Under Partial Connections for the wall plate, select the **AC power cable**.
 3. From the Selected Item pane, drag the **AC Power Connector (Female)** to the power supply port.
 4. On the power supply, flip the **power supply switch** to the on position.
 5. Above the computer, select the change view icon, then select **Front**.
 6. Select the **power button** to start the computer and verify that the computer boots successfully.